David Camilo Navarro Saiz

Senior AI Engineer | Evaluation & Research Engineering

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Summary

Senior AI Engineer with 7+ years designing and delivering AI systems across production LLM agents, applied machine learning, and computer vision. Built agent and RAG evaluation infrastructure combining synthetic QA datasets, retrieval metrics, LLM-based assessment, human review, and experiment observability. Developed reproducible benchmarks for AI coding tools integrating deterministic validation, CI evidence, and rubric-based evaluation. Combines experimental ML engineering with production Python/TypeScript systems, AWS/Azure expertise, and technical leadership, including leading a 10-engineer team delivering 30+ industrial computer-vision projects.

Skills

Evaluation & Research	RAGAS, DeepEval, LLM-as-a-Judge, human evaluation rubrics, synthetic QA datasets, retrieval metrics, experiment tracking, error analysis
AI Agents & RAG	LangGraph, LangChain, PydanticAI, LlamaIndex, retrieval-augmented generation, MCP, tool calling, agent orchestration, policy validation
Machine Learning	PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, Computer Vision, feature engineering, MLflow, DVC, CVAT
Programming & Data	Python, TypeScript, JavaScript, SQL, FastAPI, Node.js, React, Next.js, PostgreSQL, C++
Cloud & MLOps	AWS Bedrock, AgentCore, ECS/Fargate, RDS, Azure AKS, Container Apps, App Service, Azure AI Search, Docker, Kubernetes, Terraform, CI/CD
Languages	English (Fluent), Spanish (Native)

Technical Experience

Svitla Systems

Aug 2025 — Present

Sr. ML/AI Engineer

Roadie — Agentic AI Platform

Nov 2025 — Present

- Built an end-to-end evaluation framework for LLM agents and RAG systems, integrating synthetic QA generation, configurable experiments, RAGAS, DeepEval, retrieval ranking metrics, and experiment tracking to investigate retrieval and response failures.
- Developed trace-based evaluation and human-review workflows that connect retrieved context, agent decisions, tool execution results, and generated responses, enabling systematic analysis of agent behavior and reproducible scenario testing.
- Designed layered validation mechanisms for sensitive agent operations, combining deterministic business-policy checks, workflow-state verification, tool-result evidence, and LLM response adjudication to identify unsupported action-completion claims.
- Architected and delivered a multilingual, policy-governed LangGraph pickup-support agent to production, integrating controlled state transitions, confirmation gates, authorized tool execution, and structured workflow outcomes.
- Engineered shared agent-runtime reliability mechanisms, including typed execution results, operation-level timeouts, retry classification, and predictable failure handling, supporting more consistent behavior and testability.

360Training — Computer Vision & ML

Aug 2025 — Nov 2025

- Designed a multi-signal proctoring pipeline combining face detection, gaze estimation, head-pose analysis, and temporal behavioral features to identify anomalous examination activity across different visual conditions.
- Developed feature-engineering and model evaluation workflows comparing XGBoost, LightGBM, Random Forest, logistic regression, and neural models, analyzing predictive behavior and robustness across scenarios.
- Established reproducible ML experimentation using CVAT for annotation, DVC for dataset versioning, and MLflow for experiment tracking; integrated inference and heuristic services into application workflows.

Kmeleon

Jan 2025 — Aug 2025

Generative AI Engineer

- Architected a configurable multi-agent platform using LangChain and PydanticAI, supporting conversational memory, tool-driven reasoning, and collaborative agent workflows for enterprise applications.
- Designed reusable Model Context Protocol (MCP) integrations connecting agents to external services and Microsoft Teams, enabling structured tool access across different conversational workflows.
- Developed modular RAG and agent configurations integrated with Azure-oriented infrastructure, supporting maintainable retrieval and conversational AI applications.

Southdesk

Apr 2024 — Jan 2025

Machine Learning Engineer

- Engineered RAG-powered applications combining LLMs, vector retrieval, and GoHighLevel CRM integration to support knowledge-grounded business automation, customer onboarding, and personalized communication.
- Developed web-scraping and structured data-extraction pipelines for prospect analysis and CRM automation, improving the availability of data for lead-generation and outreach workflows.

El Sol — Neighborhood Educational Center

Jun 2023 — Apr 2024

Machine Learning Team Leader

- Led development of conversational AI and document-intelligence systems combining multi-format ingestion, RAG, retrieval filtering, and reranking using LangChain, FastAPI, React/TypeScript, and Node.js.
- Engineered document-driven automation and speech-enabled workflows using Azure AI Speech, supporting intelligent form completion, personalized reports, and context-aware recommendations.
- Architected containerized AI services deployed through Azure Kubernetes Service (AKS), App Service, and Container Apps, integrating CI/CD workflows to support repeatable delivery and model-driven application updates.

INTECOL SAS

May 2019 — Apr 2023

Computer Vision Engineer & Team Leader

- Led a team of 10 engineers delivering 30+ industrial computer-vision projects, directing technical architecture, model integration, project execution, budgets, and cross-functional delivery.
- Designed and implemented CNN- and YOLO-based machine-vision solutions using PyTorch, TensorFlow, Python, and C++, translating applied ML models into operational industrial systems.
- Developed computer-vision applications for object detection and human-behavior analysis, integrating model inference into desktop applications and practical monitoring environments.

Evaluation & Research Projects

Parser AI (Parser Digital) — Coding-Agent Evaluation & Enterprise AI Platform

AI Coding-Agent Benchmarking: Designed and implemented a reproducible evaluation workbench for Cursor, Claude Code, Codex, GitHub Copilot, and Gravity. Standardized full-stack development tasks, versioned prompts, and 37 validation checks, collecting code diffs, execution transcripts, CI results, security checks, timing, token usage, and human interventions. Combined deterministic checks, LLM-as-a-Judge assessments, and human scoring rubrics to compare functional correctness, code quality, maintainability, and agent autonomy through a comparative leaderboard.

AI Platform Engineering: Engineered AWS infrastructure for a LibreChat-based enterprise AI platform, validating RAG persistence migration to Amazon RDS PostgreSQL/pgvector with database access controls and S3 integration. Built authenticated AI-generated application previews using Sandpack, ECS/Fargate, HMAC-based authorization, and ALB routing.

Professional Cases — Multimodal Case Generation & AI Assessment

Agentic Case Generation: Architecting a scalable, domain-agnostic platform that transforms professional documentation, images, and specialized knowledge into realistic case studies. Designing agent-driven research and retrieval workflows to create scenarios that challenge professional reasoning and decision-making.

Knowledge Coverage & Evaluation: Developing semantic clustering and knowledge-diversity strategies to broaden topic coverage, vary case complexity, and reduce repetitive scenarios. Designing rubric-based LLM assessment to evaluate user reasoning, identify knowledge gaps, and generate structured, personalized feedback. The platform architecture uses Next.js, FastAPI, PostgreSQL/pgvector, and Redis.

Applied AI Research — Computational Imaging & Deep Learning

Contributed to research at Universidad Nacional de Colombia on CNN-based holographic image reconstruction and computational imaging. Led development of an experimental thermography-based injury-detection system. For COVID-RX, trained and compared PyTorch CNN architectures, including ResNet, DenseNet, and EfficientNet, using 36,000+ publicly sourced chest X-ray images for COVID-19 and pneumonia classification experiments.

Pupil Tracking Research — SICC 2019

Presented applied research on pupil localization and tracking using CNN-based image analysis and machine-vision techniques. Explored extracting pupil position and movement information from visual data, with applications in gaze-based interaction and educational technologies.

Computer Vision Co-Instructor — Universidad Nacional de Colombia

Collaborated with INTECOL SAS to design and deliver technical training in image processing, deep learning, and applied computer vision, connecting ML concepts with practical engineering problems.

Education

B.Sc. in Physical Engineering
Universidad Nacional de Colombia, Medellin

Selected Professional Development

Machine Learning & MLOps: MLOps Specialization (Duke University / Coursera); DS4A – Data Science for All (Correlation One / MinTIC).

Deep Learning & NLP: DeepLearning.AI coursework in Deep Learning, TensorFlow, and GANs; Natural Language Processing Specialization (Coursera).

LLM Systems & Agent Engineering: Specialized coursework in LangGraph, LLM Engineering, LangChain, and AI agent development.

Current Learning Focus: AWS Solutions Architect – Associate certification preparation and advanced AI system architecture.